

# Trade-offs in Partition Assignment Policy intended for Distributed Deployment of Dynamic Graph Embedding Systems

Stefan Nožinić

## Abstract

This paper studies trade-offs in partition assignment policy where partitioning balance and dynamic embedding quality are considered as two competing objectives. We describe a buffered event-processing pipeline in which incoming graph events are batched before a neighbor-based partitioner, with a tunable replication factor and capacity penalty, assigns vertices to partitions; embeddings are then maintained using a buffered variant of dynnode2vec. Using five real-world dynamic graph datasets (CITeseer, DBLP, AstroPh, AS-Oregon, Enron), we empirically evaluate how the number of partitions and the replication factor affect embedding quality, partition balance, edge cut, and repartitioning stability over time. Results show that the quality loss caused by partitioning is strongly dataset-dependent: embedding quality degrades as the partition count grows on two of the five datasets (AstroPh clearly, AS-Oregon more mildly), while on the other three (CITeseer, DBLP, Enron) it is stable or improves over most of the partition range, with DBLP falling slightly below its single-partition baseline only at the largest partition count tested. Among the dataset properties considered (density, average clustering coefficient, modularity), graph density correlates most closely with this split, though no single property fully explains it. In a preliminary, single-configuration experiment, increasing the replication factor substantially recovered embedding quality lost to early, uncertain partition assignment, at the cost of additional computation - a promising but not yet systematically validated direction for adding flexibility to partitioning strategies.

## Introduction

A graph is a mathematical structure consisting of vertices (or nodes) connected by edges. Graphs are widely used to model relationships and interactions in various domains (Van Der Hofstad 2024), such as social networks (Leskovec et al. 2010b) (Backstrom et al. 2006) (Rozemberczki and Sarkar 2021), collaboration networks (Savić et al. 2017), terrorist networks (Krebs 2002) and blog citation networks (Adamic and Glance 2005). In these applications, the relationships between entities can be represented as edges connecting the corresponding vertices.

In many real-world graphs, which model real-world phenomena, there are several properties that are often observed (Watts and Strogatz 1998) (Zachary 1977) (Albert and Barabási 2002):

- **Small-world property:** Most pairs of vertices can be connected by a short path, even in large graphs. This is often referred to as the “six degrees of separation” phenomenon. (Watts and Strogatz 1998)
- **Community structure:** Vertices tend to form clusters or communities, where vertices within the same community are more densely connected than those in different communities. This property is prevalent in social networks, where groups of friends or colleagues often form tightly-knit communities. (Leskovec et al. 2009)
- **Scale-free property:** The degree distribution of the graph follows a power-law, meaning that a few vertices have a very high degree, while most vertices have a low degree. This is often observed in social networks, where a small number of individuals (e.g., celebrities) have many connections, while the majority of users have relatively few connections. (Barabási and Albert 1999)

Graph vertex embeddings are a powerful technique for representing vertices in a graph as low-dimensional vectors, enabling various machine learning tasks such as vertex classification, link prediction (Leskovec et al. 2010a), and community detection. The effectiveness of these embeddings often depends on the underlying graph structure and the methods used to generate them. When faced with large graphs, the challenge of efficiently computing these embeddings while preserving the graph’s structural properties becomes paramount. Additionally, most real-world graphs are dynamic, with vertices and edges being added or removed over time. This dynamic nature introduces additional challenges in maintaining accurate and up-to-date embeddings, as the graph’s structure evolves.

Motivation to use distributed systems for graph embedding arises from the need to handle large-scale graphs that cannot fit into the memory of a single machine. Distributed systems allow for parallel processing and storage of graph data across multiple machines. However, distributing the graph across multiple machines introduces new challenges, particularly in terms of partitioning the graph and maintaining embedding quality.

When a distributed system assigns newly arriving vertices to partitions before their full neighborhood is known, it faces a practical trade-off: using more partitions improves scalability but can harm embedding quality and partition balance, while hedging an uncertain assignment by replicating a vertex across multiple partitions can recover embedding quality at the cost of additional computation. Despite its practical relevance, this trade-off has not been systematically studied in the context of dynamic graph embedding. This paper aims to fill this gap by empirically studying how partition assignment parameters, specifically the number of partitions and the replication factor, affect embedding quality, partition balance, and computational cost in distributed dynamic graph embedding systems.

## Problem Formulation

Given a dynamic graph where vertices arrive online and a distributed dynamic dynnode2vec-style (Mahdavi et al. 2018) embedding model is maintained, we want to know how partitions should be assigned to new vertices to balance:

- embedding quality,
- partition balance,
- computational cost of maintaining the assignment

## Related Work

Graph vertex embedding is a well-studied area, with various methods proposed to generate low-dimensional representations of vertices in a graph. State of the art method which is widely used is Node2Vec (Grover and Leskovec 2016) which uses random walks to capture the local and global structure of the graph. Node2Vec generates embeddings by performing biased random walks on the graph, allowing it to explore both local and global structures. The method has been shown to be effective in capturing community structures and generating meaningful embeddings for various machine learning tasks. As its improvement, DistGER (Fang et al. 2023) is a distributed graph embedding method that extends Node2Vec by leveraging distributed computing to handle large graphs. DistGER uses a similar random walk approach but optimizes walk sampling in order to maximize the information gain when selecting the next vertex to visit.

Another approach to scale Node2Vec is proposed in (Lombardo and Poggi 2019) which is based on actor model and uses a distributed framework to generate embeddings for large graphs. This method allows for parallel processing of random walks, significantly improving the efficiency of embedding generation in terms of time and resource usage.

The common ground for these methods is that they generate walks which are later used to train Word2Vec model (Church 2017) commonly used for generating embeddings in natural language processing tasks.

During the learning process, there are several state of the art approaches to parallelize the training of word2vec model. Commonly used approach in distributed environment is ensemble learning (Ji and Ling 2007) which combines multiple smaller models to create a larger model. Final model is created by aggregating smaller models using parameter server architecture (Li et al. 2013).

All enumerated methods are aimed for static graphs, where the graph structure does not change over time. However, in many real-world applications, graphs are dynamic and change over time, requiring dynamic graph embedding methods that can adapt to changes in the graph structure.

The main challenge to address the problem of distributed graph vertex embedding is partitioning the graph in a way that preserves the community structure while

ensuring that the partitions are balanced and can be processed efficiently in a distributed environment. Up until recently, most partitioning methods covered only small graphs or graphs without inherent community structure, like in (Benlic and Hao 2010) (Sanders and Schulz 2012) (Sanders and Schulz 2011) (Romero Ruiz and Segura 2018). The main focus of these methods is static graph partitioning meaning that the graph is partitioned once and then used for processing. See (Çatalyürek et al. 2023) for a broader, more recent survey of advances in graph and hypergraph partitioning. However, in many real-world applications, graphs are dynamic and change over time, requiring dynamic partitioning methods that can adapt to changes in the graph structure. In (Ugander and Backstrom 2013), a variant of label propagation algorithm is proposed for balancing partitions, but it lacks the ability to adapt to changes in the graph structure over time and it does not study the impact of partitioning on embedding quality.

For dynamic graph partitioning, there are several methods available in literature like (Stanton and Kliot 2012) and (Tsourakakis et al. 2014). These methods focus on partitioning dynamic graphs by considering the changes in the graph structure over time and adapting the partitioning accordingly. However, these methods have not been used in embedding applications so far, and their effectiveness in generating high-quality embeddings in a distributed environment remains an open question.

Distributed graph embedding methods, including scalable distributed approaches to static Node2Vec (Lombardo and Poggi 2019) and dynamic Node2Vec variants such as dynnode2vec (Mahdavi et al. 2018), enable scalable representation learning on large, evolving graphs. However, in dynamic settings with online node arrivals, a fundamental systems problem arises: newly arriving nodes must be assigned to partitions before sufficient structural or embedding information about their neighborhood is available, and it is unclear how the number of partitions and the degree of assignment redundancy affect the resulting embedding quality and partition balance.

Most existing distributed embedding systems assume either:

- static partitioning (Benlic and Hao 2010) (Sanders and Schulz 2012), where the graph is partitioned once ahead of time, or
- a batch label-propagation-style assignment such as (Ugander and Backstrom 2013), which, although not designed for online node arrivals, illustrates the same underlying limitation: each vertex is assigned a single partition without a mechanism to hedge an uncertain assignment.

## Contributions

This paper makes the following contributions:

- We propose a systematic empirical study of how partition count affects

partition assignment quality for online vertex arrivals in dynamic graph embedding systems, evaluated against a uniform-random partitioning baseline, together with a preliminary exploration of the replication factor as an additional lever.

- We evaluate the impact of partition count on embedding quality, partition balance, and edge cut, and provide preliminary, single-configuration evidence that increasing the replication factor can recover embedding quality lost to early, uncertain assignment at the cost of additional computation.
- We provide empirical evidence and analysis of this quality/computation trade-off, contributing to the understanding of how partition count, and preliminarily replication, affect partitioning in distributed dynamic graph embedding systems.

## Paper Organization

The rest of the paper is organized as follows: First, system overview is presented, then graph partitioning and embedding model are described. Next, benchmarks are explained in detail. Finally, results and discussion are presented, followed by the conclusion and references.

## Methods

### System overview

Temporal graph is represented as a sequence of events. Each event is a tuple  $(t, u, v)$  where  $t$  is the timestamp of the event and  $u$  and  $v$  are the vertices involved in the event. The events are processed in chronological order, and the graph is updated accordingly. The system maintains a distributed dynamic Node2Vec-style embedding model, which is updated as new events arrive. The partition assignment strategy determines how new vertices are assigned to partitions in the distributed system.

The system buffers incoming events for a short period before assigning them to partitions. This buffering allows the system to make more informed partitioning decisions by considering a batch of events together, rather than assigning partitions immediately upon the arrival of each event. Once the events are buffered, the partition assignment strategy is applied, and the embeddings are updated accordingly.

In the following listing, pseudocode for the system is presented:

```
global state buffer = {}

when new event (t, u, v) arrives:
    buffer.add((t, u, v))
```

```

if buffer.size() >= MAX_BUFFER_SIZE:
    assign_partitions(buffer)
    update_embeddings(buffer)
    buffer.clear()

```

Buffer also defines new graph snapshot. Let  $G_n$  be the graph snapshot after processing the first  $n$  buffers. Now, let  $B_n$  be the  $n$ -th buffer of events. The graph snapshot  $G_{n+1}$  is obtained by applying the events in buffer  $B_n$  to the graph snapshot  $G_n$ . Specifically,  $G_{n+1} = G_n \cup B_n$ , where the union operation adds the new vertices and edges from the events in  $B_n$  to the existing graph snapshot  $G_n$ . This process allows the system to maintain an up-to-date representation of the dynamic graph as new events are processed.

## Graph partitioning

Let  $\mathcal{P} = \{p_1, \dots, p_P\}$  denote the current partitioning of the graph into  $P$  partitions. The partition of a vertex is the partition with the highest score. The partitioning strategy assigns the vertex to the partition that contains the most neighbors of the vertex in the buffer, while keeping the partition sizes balanced. The partitioner also has a replication factor, which allows it to assign a vertex to multiple partitions. The partitioner has a capacity penalty, which penalizes partitions that have more vertices than the average partition size, to encourage more balanced partitions.

Formally, the score of partition  $p_i \in \mathcal{P}$  for a vertex  $v$  is defined as:

$$S(p_i, v) = N(p_i, v) - \mu \cdot \max \left( 0, |p_i| - \alpha \cdot (1 + \epsilon) \cdot \frac{1}{P} \sum_{p_j \in \mathcal{P}} |p_j| \right)$$

where  $N(p_i, v)$  is the number of neighbors of vertex  $v$  in partition  $p_i$  in the buffer,  $\mu$  is the capacity penalty coefficient,  $\alpha$  is the weight of the average partition size in the capacity penalty,  $\epsilon$  is the imbalance tolerance, and  $|p_i|$  is the size of partition  $p_i$ . The partitioner assigns the vertex to the top  $RF$  partitions of  $\mathcal{P}$  with the highest scores, where  $RF$  is the replication factor. If there are multiple partitions with same scores, the partitioner randomly assigns the vertex to some of those partitions until it reaches the replication factor.

Because the partitioner only considers neighbors observed within the current buffer, its per-buffer cost is bounded independently of the total graph size processed so far. Let  $B$  denote the buffer size and let  $\deg_B(v)$  denote the number of buffer-local edge endpoints incident to vertex  $v$ , so that  $\sum_v \deg_B(v) \leq 2B$ . For each buffered edge, looking up the current partition membership of its endpoints and incrementing the corresponding neighbor-count tallies  $N(p_i, v)$  takes  $O(1)$  amortized time per edge endpoint per replica, giving  $O(RF \cdot B)$  total work to compute all neighbor counts for the buffer, since a vertex can belong to up to  $RF$  partitions. The average partition size term  $\frac{1}{P} \sum_{p_j \in \mathcal{P}} |p_j|$  is maintained

incrementally and reused across all vertices in the buffer, costing  $O(P)$  per buffer rather than per vertex. For each of the at most  $2B$  vertices touched by the buffer, computing  $S(p_i, v)$  for all  $P$  partitions and selecting the top  $RF$  scores costs  $O(P)$  per vertex. The total cost of partitioning one buffer is therefore

$$O(B \cdot (P + RF))$$

which, for fixed  $P$  and  $RF$ , is  $O(B)$ : linear in the buffer size and independent of the total number of vertices or edges in the graph processed so far. Over a stream of  $T$  buffers covering  $N = T \cdot B$  events in total, the cumulative partitioning cost is  $O(N \cdot (P + RF))$ , i.e., linear in the total number of events for the fixed, small values  $P \in \{1, 2, 4, 8\}$  and  $RF \in \{1, 3\}$  used in this study.

This bound is asymptotically dominated by the cost of the embedding step for the same buffer. `dynnode2vec` generates `n_walks` random walks of length `walk_size` from each of the (up to  $2B$ ) evolving vertices, and trains skip-gram over the resulting corpus for several epochs, at a cost of  $O(|\Delta V_t| \cdot n\_walks \cdot walk\_size \cdot window\_size \cdot epochs)$  per partition replica. Both costs are linear in the number of vertices touched by the buffer, but the constant factor for embedding ( $walks \times walk\ length \times window\ size \times epochs$ ) is much larger than partitioning’s constant factor ( $P + RF \leq 11$  in this study), which is why partitioning time is negligible compared to embedding time in practice.

## Embedding model

Given dynamic graph (i.e., a graph that changes over time), the `dynnode2vec` algorithm (Mahdavi et al. 2018) learns continuous feature representations for nodes in the graph at different time steps. The main idea behind `dynnode2vec` is to extend the `node2vec` algorithm (Grover and Leskovec 2016) to handle dynamic graphs by incorporating temporal information into the random walk strategy and embedding learning process. The goal is to capture both the structural and temporal dynamics of the graph in the learned embeddings.

In the following listing, pseudocode for the `dynnode2vec` algorithm is presented:

Input: Graph snapshots `G1, G2, ..., GT`

Output: Embeddings `Z1, Z2, ..., ZT`

```
// t = 1
```

```
Run static node2vec on G1 to train Skip-gram_1 and obtain Z1.
```

```
For t = 2 ... T:
```

```
    // Identify evolving nodes between G_{t-1} and G_t
```

```
    Compute:
```

```
        V_add = nodes added from t-1 to t
```

```
        E_add = edges added from t-1 to t
```

```
        E_del = edges deleted from t-1 to t
```

```

delta_V_t = V_add union { v in V_t | exists (v, u) in (E_add union E_del) }

// Evolving random walks only from changed regions
Walk_n = node2vec_walks(G_t, start_nodes=delta_V_t, p, q, walk_length, num_walks)

// Dynamic Skip-gram: initialize from previous time
Initialize Skip-gram_t with weights of Skip-gram_{t-1}
Update Skip-gram_t vocabulary for any new vertices
Train Skip-gram_t on Walk_n
Extract Z_t from Skip-gram_t
End For

```

In this paper, graph is represented as a sequence of events. Algorithm can be easily adapted to this representation by buffering events and applying them to the graph snapshot at each time step. The algorithm captures the evolving nature of the graph by focusing on the changed regions and updating the embeddings accordingly.

Since one vertex can be assigned to multiple partitions, each partition produces one embedding for corresponding vertex. Let  $z_p = h_p(u)$  be the embedding of vertex  $u$  in partition  $p$ . The final embedding of vertex  $u$  is obtained by averaging the embeddings from all partitions that contain the vertex:

$$z_u = \frac{1}{|P_u|} \sum_{p \in P_u} z_p$$

where  $P_u = \{p | u \in V_p\}$  is the set of partitions that contain vertex  $u$ . This averaging process allows the system to combine information from multiple partitions, potentially improving the quality of the final embedding by leveraging diverse perspectives on the vertex's relationships within the graph. This step requires communication between partitions to share the embeddings of vertices that are replicated across multiple partitions. The communication overhead is proportional to the number of replicated vertices and the size of their embeddings.

If replication factor is set to 1, then the final embedding of vertex  $u$  is simply the embedding from the single partition that contains the vertex:

$$z_u = z_p$$

where  $p$  is the partition that contains vertex  $u$ . In this case, the embedding is solely based on the information available in that single partition, which may limit the quality of the embedding if the partition does not capture sufficient context about the vertex's relationships within the graph.

## Benchmarks and evaluation metrics

To evaluate the effectiveness of the graph vertex embeddings generated using community-aware partitioning, several criteria are considered:

**Embedding Quality:** The quality of the embeddings is assessed using F1-score of reconstructed graph (Yip et al. 2023), which measures how well the embeddings capture the relationships between vertices in the original graph. Higher F1-scores indicate better preservation of graph structure in the embeddings. Let  $G = (V, E)$  be the original graph and  $G' = (V, E')$  be the reconstructed graph from embeddings.  $G'$  is constructed by connecting the  $m$  closest vertex pairs in the embedding space, where  $m$  is the number of edges in the original graph. The F1-score is calculated as follows:

$$F1 = 2 * \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

where

$$\text{Precision} = \frac{1}{|V|} \sum_{v \in V} \frac{|N(v) \cap N'(v)|}{|N'(v)|}$$

$$\text{Recall} = \frac{1}{|V|} \sum_{v \in V} \frac{|N(v) \cap N'(v)|}{|N(v)|}$$

and  $N(v)$  and  $N'(v)$  are the neighbors of vertex  $v$  in the original and reconstructed graphs, respectively.

**Partition Quality:** The quality of the partitions is evaluated using edge cut, which measures the number of edges that connect vertices in different partitions. Lower edge cuts indicate better preservation of community structures within partitions. Edge cut is defined as:

$$\text{EdgeCut} = \frac{1}{|E|} \sum_{(u,v) \in E} \mathbb{I}(p(u) \neq p(v))$$

where  $E$  is the set of edges in the original graph,  $p(u)$  is the partition of vertex  $u$ , and  $\mathbb{I}(\cdot)$  is the indicator function.

**Balance of Partitions:** The balance of the partitions is assessed by measuring the size of each partition and ensuring that they are within a bounded interval. This helps to ensure that the workload is evenly distributed across machines in the distributed environment. Formally, balance is defined as:

$$B = 1 + \frac{1}{K} \sum_{i=1}^K \frac{|s_i - \bar{s}|}{\bar{s}}$$

where  $K$  is the number of partitions,  $s_i$  is the number of vertices assigned to partition  $i$ , and  $\bar{s} = \frac{1}{K} \sum_{i=1}^K s_i$  is the average partition size. A perfectly balanced partitioning yields  $B = 1$ ; higher values indicate greater imbalance.

**Partitioning Time:** The time taken to partition the graph is measured to evaluate the efficiency of the partitioning algorithms. Faster partitioning times are preferred, especially for large graphs, as they reduce the overall processing time in a distributed environment. In this paper, partitioning time is analyzed asymptotically via the per-buffer complexity bound derived in the Graph partitioning section, and it is measured directly as wall-clock partitioning and embedding time as a function of buffer size for CITESEER and AstroPh, in the Buffer size sensitivity analysis section.

**Repartitioning amount:** The amount of repartitioning required when new vertices arrive is measured to assess the stability of the partitioning strategy. Lower amounts of repartitioning indicate that the partitioning strategy is more stable and can handle dynamic changes in the graph without significant disruption. Given two graph snapshots  $G_n$  and  $G_{n+1}$ , the amount of repartitioning is defined as:

$$\text{RepartitioningAmount} = \frac{1}{|V_n|} \sum_{v \in V_n} \mathbb{I}(p_n(v) \neq p_{n+1}(v))$$

Where  $V_n$  is the set of vertices in graph snapshot  $G_n$ ,  $p_n(v)$  is the partition of vertex  $v$  in snapshot  $G_n$ , and  $p_{n+1}(v)$  is the partition of vertex  $v$  in snapshot  $G_{n+1}$ . This metric quantifies the proportion of vertices that have been reassigned to different partitions between two consecutive snapshots, providing insight into the stability and adaptability of the partitioning strategy in response to dynamic changes in the graph. It has to be noted that only vertices that are present in both snapshots are considered for this metric, as newly added vertices do not have a previous partition assignment to compare against and thus do not contribute to the measure of repartitioning amount.

## Results and discussion

### Partitioning hyperparameters

Throughout the experiments, the following hyperparameters are used for partitioning:

Table 1: Partitioning hyperparameters used throughout the experiments.

| Hyperparameter | Value |
|----------------|-------|
| Buffer size    | 1000  |

| Hyperparameter          | Value        |
|-------------------------|--------------|
| $\mu$                   | 1            |
| $\alpha$                | 1            |
| $\epsilon$              | 0.1          |
| Partitions (P)          | {1, 2, 4, 8} |
| Replication factor (RF) | 1            |

These hyperparameters are chosen to balance the trade-offs between embedding quality, partition balance, and computational cost. In particular, the buffer size is set to 1000 to allow for sufficient information to be gathered before making partitioning decisions; this value is held fixed throughout the experiments, and its effect on embedding quality is explored as part of the buffer size sweep in the next section.

The capacity penalty coefficient ( $\mu$ ) is set to 1 to encourage balanced partitions, while the weight of the average partition size ( $\alpha$ ) is also set to 1 to ensure that the capacity penalty is proportional to the average partition size. The imbalance tolerance ( $\epsilon$ ) is set to 0.1 to allow for some flexibility in partition sizes while still encouraging balance. The number of partitions (P) is varied between 1, 2, 4, and 8 to evaluate the impact of partitioning strategy on embedding quality and partition balance. Finally, the replication factor (RF) is varied between 1 and 3 to analyze the effect of replicating vertices across multiple partitions on embedding quality.

For the embedding model itself, the following dataset-specific hyperparameters are used, obtained from a preliminary search over the node2vec return/in-out parameters  $p, q$  and the embedding dimension:

Table 2: Embedding hyperparameters (node2vec  $p, q$ , and embedding dimension) used for each dataset.

| Dataset   | p   | q    | dim |
|-----------|-----|------|-----|
| CITeseer  | 0.5 | 0.25 | 50  |
| DBLP      | 0.5 | 1    | 25  |
| AstroPh   | 2   | 0.25 | 50  |
| AS-Oregon | 0.5 | 2    | 128 |
| Enron     | 0.5 | 1    | 128 |

## Datasets

Experiments are performed on five real-world graph datasets, summarized in the following table.

Table 3: Datasets used in the experiments.

| Dataset   | Nodes | Edges  | Average degree | Average cluster-ing coefficient | Density | Modularity |
|-----------|-------|--------|----------------|---------------------------------|---------|------------|
| CITeseer  | 3264  | 4536   | 2.78           | 0.145                           | 0.0009  | 0.72       |
| DBLP      | 17716 | 52867  | 5.97           | 0.134                           | 0.00034 | 0.58       |
| AstroPh   | 18772 | 198110 | 21.11          | 0.631                           | 0.00112 | 0.33       |
| AS-Oregon | 11806 | 38781  | 6.57           | 0.399                           | 0.00056 | 0.57       |
| Enron     | 36692 | 183831 | 10.02          | 0.497                           | 0.00027 | 0.36       |

CITeseer is a citation network of scientific publications. DBLP and AstroPh are collaboration/citation networks derived from SNAP, where AstroPh links co-authors of astrophysics papers. AS-Oregon is a network of autonomous systems and their peering connections. Enron is an email communication network in which vertices are email addresses and edges represent messages exchanged between them. These datasets vary in size and density, which allows the effect of partitioning to be observed under different community structures.

As a static, non-distributed reference point, running node2vec on the full, unpartitioned graph gives the following F1 reconstruction scores:

Table 4: Average F1 reconstruction score of static node2vec on the full, unpartitioned graph for each dataset.

| Dataset   | Average F1 score |
|-----------|------------------|
| CITeseer  | 34.82%           |
| DBLP      | 60.3%            |
| AstroPh   | 70.41%           |
| AS-Oregon | 31.26%           |
| Enron     | 20.42%           |

These values serve as a rough upper bound against which the effect of partitioning, buffering, and dynamic re-embedding can be compared: a system that partitions the graph and processes it incrementally is generally not expected to exceed this quality, since it operates with a fraction of the graph’s global information at any given time and adapts to it incrementally rather than embedding the full graph at once. CITeseer is a notable exception to this expectation, discussed in the next section: the buffered dynnode2vec pipeline exceeds the static baseline there even without partitioning, plausibly because repeated incremental re-embedding

across buffers gives the model more effective training signal than a single static pass over the graph.

### Sensitivity analysis of the capacity penalty parameter

The most common neighbor partitioner uses a capacity penalty coefficient  $\mu$  to discourage assigning vertices to partitions that are already larger than average. This section analyzes the effect of  $\mu$  on partition balance, varying  $\mu \in \{0, 0.5, 1, 1.5, 2\}$  across all five datasets with 2 and 4 partitions.  $\mu = 0$  disables the penalty entirely, so balance is determined purely by neighbor counts; higher  $\mu$  increasingly rebalances vertex placement at the expense of prioritizing neighbor locality. The other side of this trade-off — the F1 cost of increasing  $\mu$  — is quantified from a preliminary hyperparameter search (buffer size 1000,  $RF = 1$ ) conducted on CITESEER and AstroPh over  $\mu \in \{0, 0.5, 1\}$ :

Table 5: F1 reconstruction score vs.  $\mu$  and partition count for AstroPh (buffer size 1000,  $RF = 1$ ).

| P | $\mu = 0$ | $\mu = 0.5$ | $\mu = 1$ |
|---|-----------|-------------|-----------|
| 1 | 57.00%    | 57.00%      | 57.00%    |
| 2 | 57.00%    | 54.82%      | 56.28%    |
| 4 | 57.00%    | 52.00%      | 49.92%    |
| 8 | 57.00%    | 33.93%      | 46.61%    |

Table 6: F1 reconstruction score vs.  $\mu$  and partition count for CITESEER (buffer size 1000,  $RF = 1$ ).

| P | $\mu = 0$ | $\mu = 0.5$ | $\mu = 1$ |
|---|-----------|-------------|-----------|
| 1 | 58.00%    | 58.00%      | 58.00%    |
| 2 | 58.00%    | 41.76%      | 46.18%    |
| 4 | 58.00%    | 43.54%      | 44.35%    |
| 8 | 58.00%    | 44.31%      | 49.09%    |

The  $\mu = 0$  column is flat at the  $P = 1$  value for every  $P$ : with the capacity penalty disabled, the partitioner’s assignment is dominated by whichever partition first pulls ahead in neighbor count, so it is not really testing partitioned embedding at all and its high F1 is not a meaningful “ $\mu = 0$  is best” result. Once  $\mu > 0$  forces genuine partitioning, F1 drops noticeably below that flat baseline for most  $(P, \mu)$  combinations — most sharply for AstroPh at  $\mu = 0.5$ ,  $P = 8$  (57.00%  $\rightarrow$  33.93%) — confirming that better balance is bought at a real embedding-quality cost.  $\mu = 1$  partially recovers quality relative to  $\mu = 0.5$  in most rows (though not uniformly, e.g. AstroPh  $P = 4$ ), which is consistent with  $\mu = 1$  being the trade-off point chosen for the rest of this paper’s experiments, though this preliminary

sweep covers only two datasets and  $\mu \in \{0, 0.5, 1\}$ , not the full  $\mu \in \{0, \dots, 2\}$  range used for the balance-only tables.

The tables below show the average balance at the final iteration for each combination of  $\mu$  and partition count.  $B = 1$  is a perfectly balanced partitioning; higher values indicate greater imbalance.

Table 7: Average balance at the final iteration for AS-Oregon dataset across different  $\mu$  values and partition counts.

| Penalty | P=2  | P=4  |
|---------|------|------|
| 0.0     | 2.00 | 2.50 |
| 0.5     | 1.18 | 1.49 |
| 1.0     | 1.18 | 1.20 |
| 1.5     | 1.10 | 1.21 |
| 2.0     | 1.05 | 1.19 |

Table 8: Average balance at the final iteration for AstroPh dataset across different  $\mu$  values and partition counts.

| Penalty | P=2  | P=4  |
|---------|------|------|
| 0.0     | 2.00 | 2.50 |
| 0.5     | 1.15 | 1.18 |
| 1.0     | 1.13 | 1.12 |
| 1.5     | 1.08 | 1.12 |
| 2.0     | 1.05 | 1.22 |

Table 9: Average balance at the final iteration for CITESEER dataset across different  $\mu$  values and partition counts.

| Penalty | P=2  | P=4  |
|---------|------|------|
| 0.0     | 2.00 | 2.50 |
| 0.5     | 1.61 | 1.61 |
| 1.0     | 1.31 | 1.63 |
| 1.5     | 1.17 | 1.50 |
| 2.0     | 1.17 | 1.17 |

Table 10: Average balance at the final iteration for DBLP dataset across different  $\mu$  values and partition counts.

| Penalty | P=2  | P=4  |
|---------|------|------|
| 0.0     | 2.00 | 2.50 |
| 0.5     | 1.08 | 1.16 |
| 1.0     | 1.03 | 1.09 |
| 1.5     | 1.14 | 1.20 |
| 2.0     | 1.01 | 1.14 |

Table 11: Average balance at the final iteration for Enron dataset across different  $\mu$  values and partition counts.

| Penalty | P=2  | P=4  |
|---------|------|------|
| 0.0     | 2.00 | 2.50 |
| 0.5     | 1.10 | 1.14 |
| 1.0     | 1.10 | 1.10 |
| 1.5     | 1.02 | 1.02 |
| 2.0     | 1.06 | 1.06 |

The following plots show the (smoothed) average balance over iterations for each  $\mu$  value, for  $P=2$  and  $P=4$  partitions side by side.

Across all datasets,  $\mu = 0$  (no capacity penalty) produces the worst balance: at both  $P = 2$  and  $P = 4$  the measured balance (2.00 and 2.50, identically across all five datasets) exactly matches the value obtained when every vertex collapses into a single partition, since with no capacity penalty the partition with the most buffer-local neighbors always wins and, once one partition pulls ahead early in the stream, it keeps attracting new vertices. Increasing  $\mu$  from 0 to 1 yields the largest balance improvement; beyond  $\mu = 1$ –1.5, returns diminish and balance mostly plateaus, with CITESEER remaining the hardest dataset to balance at every  $\mu$  value tested.

This  $\mu$ -sensitivity sweep was run as a separate experiment from the main  $P/RF$  results reported in the next section; consequently the  $\mu = 1$  balance values shown here can differ slightly from the  $\mu = 1$ ,  $RF = 1$  rows of the main results tables in the embedding quality, partition balance and edge cut section for the same nominal configuration (e.g. DBLP: 1.03/1.09 here vs. 1.10/1.20 there for  $P = 2/P = 4$ ) due to run-to-run variance between the two run sets; the qualitative trends — near-perfect balance for DBLP/AstroPh/AS-Oregon/Enron and a harder-to-balance CITESEER — are consistent across both.

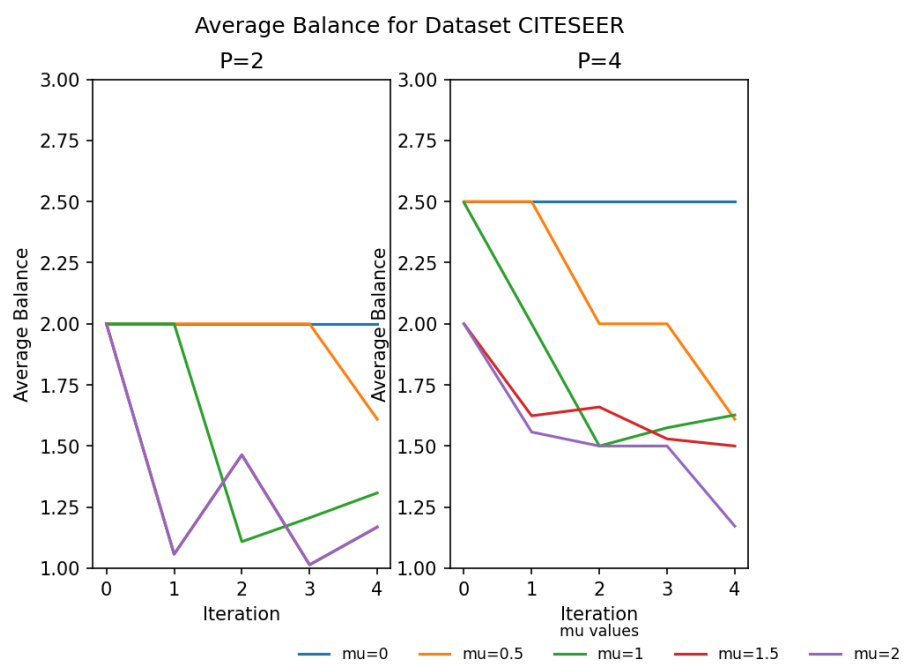


Figure 1: CITESEER Balance vs  $\mu$

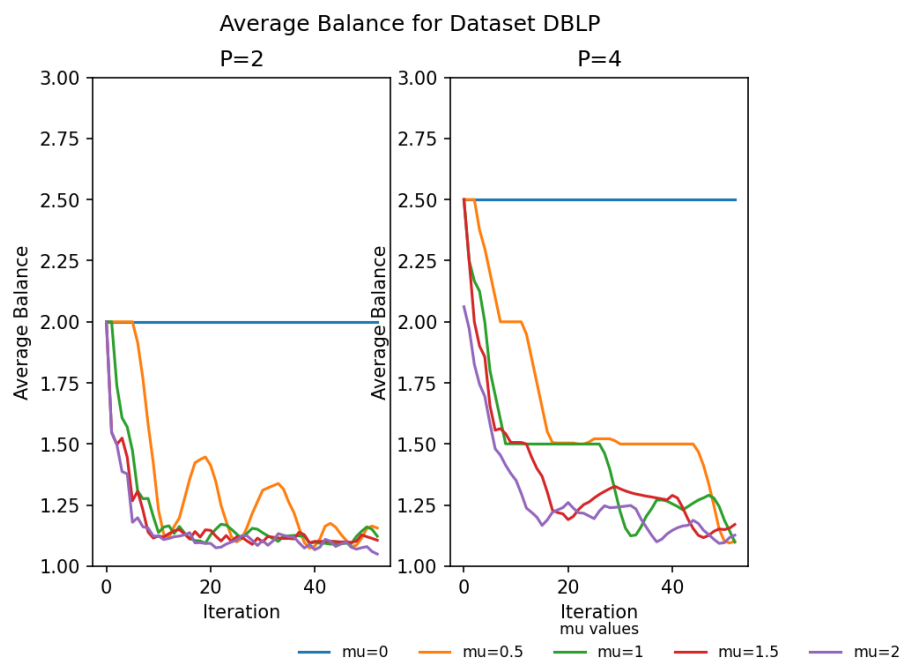


Figure 2: DBLP Balance vs mu

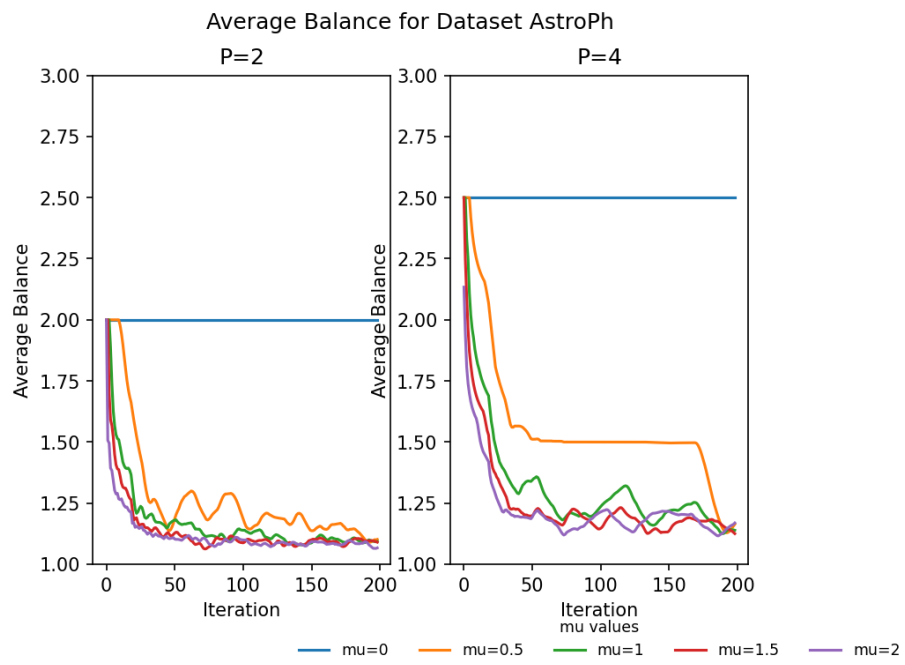


Figure 3: AstroPh Balance vs  $\mu$

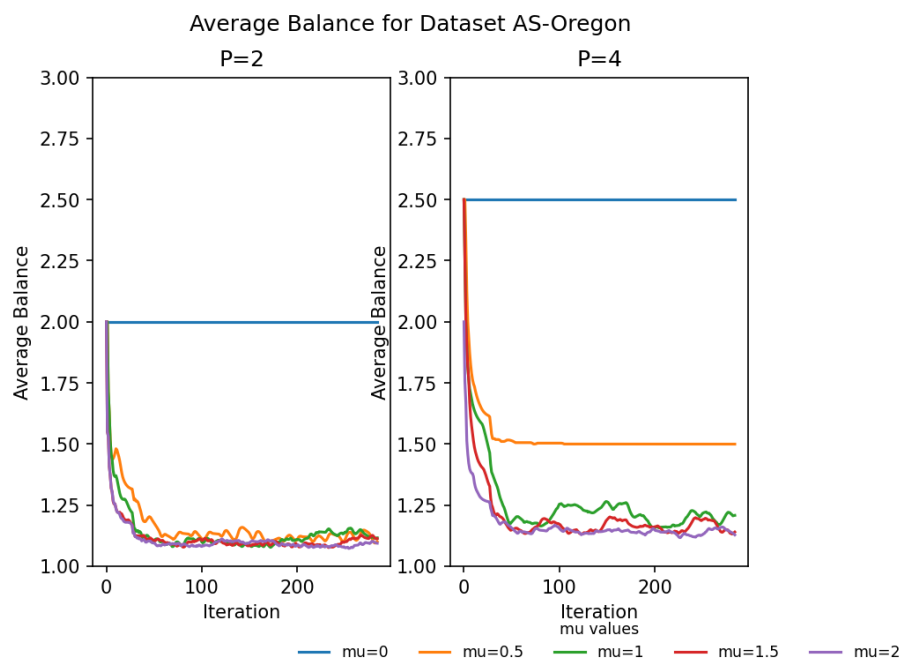


Figure 4: AS-Oregon Balance vs  $\mu$

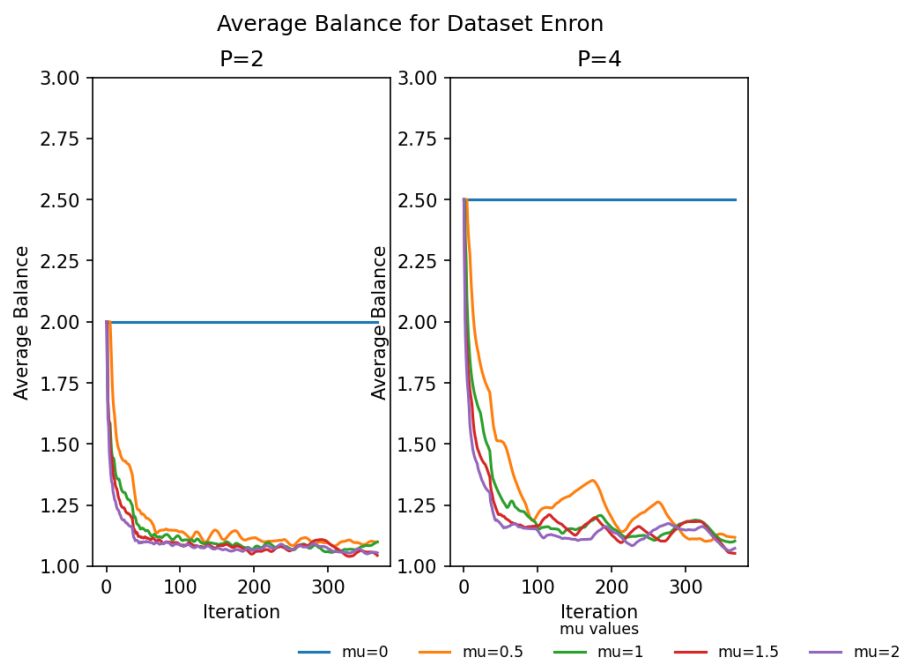


Figure 5: Enron Balance vs mu

## Buffer size sensitivity analysis

We analyzed the impact of different buffer sizes on embedding quality and on wall-clock time performance of the buffered dynnode2vec algorithm, sweeping buffer size  $\in \{100, 200, \dots, 2000\}$  at  $P = 4$  for CITESEER and AstroPh.

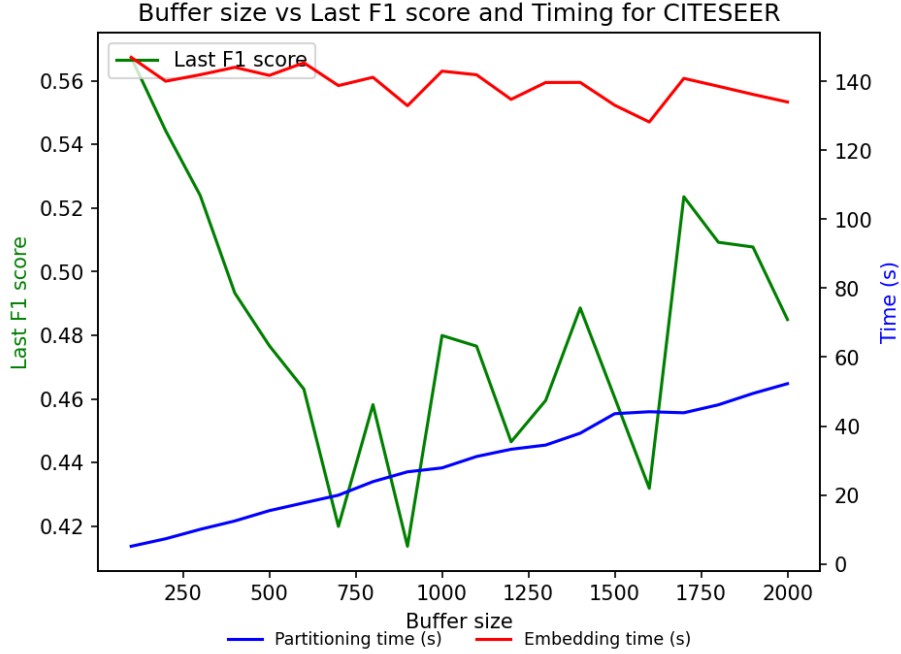


Figure 6: CITESEER: last-iteration F1 score, and wall-clock partitioning/embedding time, vs. buffer size

Each plot reports three wall-clock-derived series against buffer size: the last-iteration F1 reconstruction score (left axis), and directly measured partitioning time and embedding time, in seconds, summed over the run (right axis). The effect of buffer size on F1 is dataset-dependent rather than uniformly positive: on AstroPh, F1 rises sharply from  $\sim 0.27$  at buffer size 100 to  $\sim 0.50$  by buffer size 400 and then continues a shallow upward trend to  $\sim 0.53$  at buffer size 2000, so larger buffers clearly help. On CITESEER, the opposite pattern holds: F1 is highest at the smallest buffer size tested ( $\sim 0.56$  at 100–200) and trends down, noisily, to  $\sim 0.42$ – $0.49$  for most buffer sizes above 600, only partially recovering at a few buffer sizes (e.g.  $\sim 0.52$  at 1700) before ending at  $\sim 0.48$  at buffer size 2000 — larger buffers do not generally help CITESEER, and the relationship is closer to a shallow decline with high variance than to a monotone improvement.

Both wall-clock series show partitioning time increasing with buffer size, consistent with the  $O(B \cdot (P + RF))$  per-buffer cost, while embedding time decreases

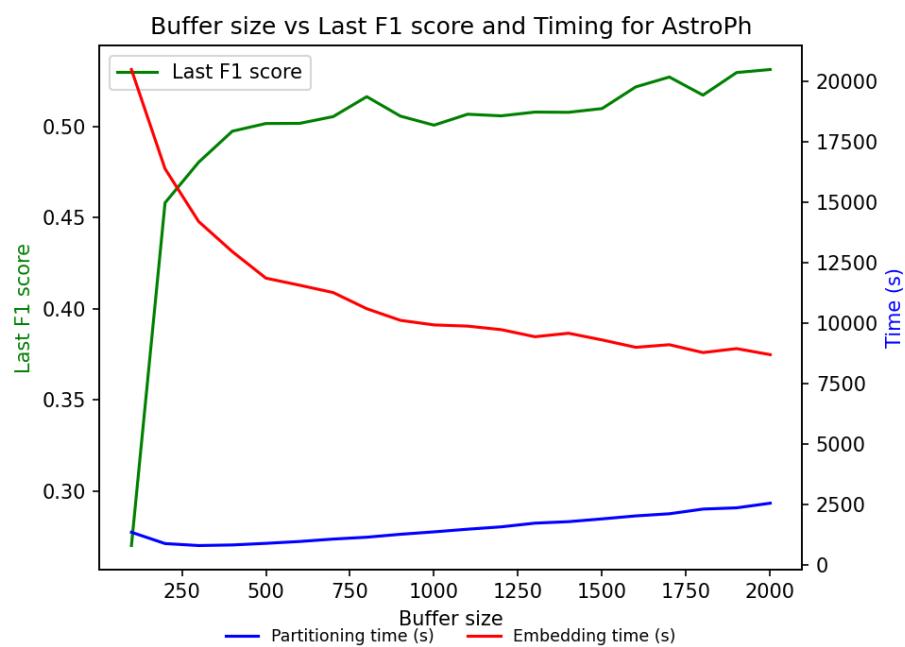


Figure 7: AstroPh: last-iteration F1 score, and wall-clock partitioning/embedding time, vs. buffer size

with buffer size on AstroPh (from  $\sim 20,500$ s at buffer 100 to  $\sim 8,700$ s at buffer 2000, as fewer, larger re-embedding steps reduce restart overhead) and stays roughly flat on CITESEER ( $\sim 130$ – $145$ s across the whole range). Partitioning time is a much smaller share of total time than embedding time at small buffer sizes (e.g. CITESEER at buffer 100:  $\sim 5$ s partitioning vs.  $\sim 145$ s embedding, a  $\sim 30\times$  gap), but that gap narrows substantially at larger buffer sizes — at buffer size 1000, the value used throughout the rest of this paper, partitioning time is roughly 20–30% of embedding time on both datasets (CITESEER:  $\sim 45$ s vs.  $\sim 140$ s; AstroPh:  $\sim 1,700$ s vs.  $\sim 10,000$ s). Partitioning time therefore remains smaller than embedding time at every buffer size tested.

### Embedding quality, partition balance and edge cut

Using the pipeline described in the system overview, the following tables report, for each dataset and number of partitions  $P \in \{1, 2, 4, 8\}$  with replication factor  $RF = 1$ , the F1 reconstruction score, the edge cut, and the balance of the resulting partitions, averaged over 10 iterations of the buffered stream.

Table 12: F1 reconstruction score, edge cut, and balance for CITESEER with buffered dynnode2vec,  $RF = 1$ .

| P | F1                   | Edge cut | Balance |
|---|----------------------|----------|---------|
| 1 | $47.57\% \pm 1.56\%$ | 0%       | 1       |
| 2 | $45.81\% \pm 0.59\%$ | 23%      | 1.31    |
| 4 | $48.06\% \pm 0.53\%$ | 39%      | 1.63    |
| 8 | $54.81\% \pm 1.66\%$ | 48%      | 1.75    |

Table 13: F1 reconstruction score, edge cut, and balance for DBLP with buffered dynnode2vec,  $RF = 1$ .

| P | F1                   | Edge cut | Balance |
|---|----------------------|----------|---------|
| 1 | $50.91\% \pm 0.31\%$ | 0%       | 1       |
| 2 | $54.11\% \pm 0.18\%$ | 32%      | 1.10    |
| 4 | $54.51\% \pm 0.15\%$ | 50%      | 1.20    |
| 8 | $49.87\% \pm 0.16\%$ | 59%      | 1.31    |

Table 14: F1 reconstruction score, edge cut, and balance for AstroPh with buffered dynnode2vec,  $RF = 1$ .

| P | F1                   | Edge cut | Balance |
|---|----------------------|----------|---------|
| 1 | $57.70\% \pm 0.11\%$ | 0%       | 1       |
| 2 | $54.67\% \pm 0.36\%$ | 35%      | 1.08    |

| P | F1                 | Edge cut | Balance |
|---|--------------------|----------|---------|
| 4 | 50.95% $\pm$ 0.45% | 56%      | 1.11    |
| 8 | 45.94% $\pm$ 0.51% | 65%      | 1.26    |

Table 15: F1 reconstruction score, edge cut, and balance for AS-Oregon with buffered dynnode2vec, RF = 1.

| P | F1                 | Edge cut | Balance |
|---|--------------------|----------|---------|
| 1 | 26.17% $\pm$ 0.57% | 0%       | 1       |
| 2 | 24.88% $\pm$ 2.72% | 30.44%   | 1.21    |
| 4 | 24.30% $\pm$ 0.98% | 53.79%   | 1.26    |
| 8 | 21.45% $\pm$ 1.19% | 71.56%   | 1.27    |

Table 16: F1 reconstruction score, edge cut, and balance for Enron with buffered dynnode2vec, RF = 1.

| P | F1                 | Edge cut | Balance |
|---|--------------------|----------|---------|
| 1 | 18.03% $\pm$ 0.44% | 0%       | 1       |
| 2 | 24.84% $\pm$ 0.38% | 33.07%   | 1.09    |
| 4 | 27.44% $\pm$ 2.83% | 49.18%   | 1.08    |
| 8 | 32.36% $\pm$ 1.24% | 61.99%   | 1.25    |

As expected, edge cut increases monotonically with the number of partitions across all datasets, since splitting the vertex set into more partitions necessarily severs more cross-community edges. Balance stays close to the ideal value of 1 for DBLP, AstroPh, AS-Oregon, and Enron (at most 1.35 even at  $P = 8$ ), confirming that the neighbor-based partitioner with capacity penalty distributes vertices close to evenly on these datasets without an explicit load-balancing step. CITESEER is a notable exception: its imbalance grows to 1.63 at  $P = 4$  and 1.75 at  $P = 8$ , indicating a substantially less even partition size distribution than on the other datasets under the same capacity penalty settings.

The effect of partitioning on embedding quality (F1), however, is dataset-dependent. On CITESEER, DBLP, and Enron, F1 score remains stable or even improves as  $P$  increases, despite the growing edge cut: this suggests that, on these graphs, most of a vertex’s relevant neighborhood is captured within its own partition, so splitting the graph does not meaningfully harm the quality of the embedding that is reconstructed from local, per-partition context.

CITESEER stands out further still: even at  $P = 1$ , its buffered F1 score (47%) already exceeds the static node2vec baseline (34.82%) reported in the datasets section, and it continues to rise with  $P$ , reaching 54.81% at  $P = 8$ .

On AstroPh and AS-Oregon, F1 is lower at  $P = 8$  than at  $P = 1$ , since a larger share of each vertex’s neighbors end up outside its assigned partition, and this information is not available to the local embedding process. AstroPh degrades monotonically with  $P$ ; AS-Oregon also decreases monotonically from  $P = 1$  through  $P = 8$  (26.17%  $\rightarrow$  24.88%  $\rightarrow$  24.30%  $\rightarrow$  21.45%), though more gently, and, per the significance analysis in the statistical significance of the partition-count effect section, only the  $P = 1$  vs  $P = 8$  gap is statistically distinguishable from run-to-run noise.

Across the five datasets, the signed F1 change from  $P = 1$  to  $P = 8$  ( $\Delta F1 = F1_{P=8} - F1_{P=1}$ ) correlates most strongly with graph density (Spearman  $\rho \approx -0.7$ ), and more weakly with modularity ( $\rho \approx +0.4$ ) and average clustering coefficient ( $\rho \approx -0.3$ ). With only five data points, none of these correlations reaches conventional statistical significance ( $p \approx 0.19$  for the density correlation), so they should be read as suggestive rather than confirmatory. Even taken at face value, none of these statistics cleanly separates the two groups with a single threshold: CITESEER, part of the stable/improving group, is denser than AS-Oregon, which degrades; and Enron combines the lowest density, the lowest modularity among the stable group, and only moderate clustering, yet shows the largest F1 improvement of any dataset in the study. This suggests density is the closest available proxy among the properties measured here, but not a complete explanation, and a larger dataset sample would be needed to identify the actual driver with confidence.

Partitioning time itself was consistently smaller than embedding time, as directly measured by wall-clock partitioning and embedding time in the buffer size sensitivity experiments in the buffer size sensitivity analysis section: at buffer size 1000, the value used throughout the P/RF sweep, partitioning time is roughly 20–30% of embedding time on both CITESEER and AstroPh, growing to a much larger margin at small buffer sizes. This is consistent with the  $O(B \cdot (P + RF))$  per-buffer complexity bound derived in the Graph partitioning section, whose constant factor is smaller than the embedding step’s; the dominant cost of processing a buffer is the embedding computation on the slowest (largest or most active) partition, though partitioning is not negligible enough to ignore entirely at the buffer sizes used here.

## Statistical significance of the partition-count effect

The trends described so far are stated in terms of mean F1 scores, but with only a few repeated runs per configuration it is worth checking whether the differences between partition counts are statistically distinguishable from run-to-run variation rather than merely eyeballed from the tables. For each dataset and partition count, the reconstruction F1 at the final iteration of every independent run is collected (between 3 and 11 runs per configuration; the per-run standard deviations are the  $\pm$  values reported in the tables in the embedding quality, partition balance and edge cut section), and partition counts are compared pairwise: each  $P$  against the single-partition baseline, and each adjacent pair

$(P, 2P)$ . Each run is a full, independent re-execution of the buffered pipeline from scratch — fresh random walks, skip-gram initialization, and partitioner tie-breaking — pooled across separate machine run sets rather than repeated iterations within a single stream, so the independence assumption behind these tests is reasonable; nonetheless, some of the effect sizes reported next are large enough that the raw percentage-point gap, not  $d$  alone, should carry the practical interpretation. Three complementary statistics are reported for every pair:

- **Welch’s two-sample  $t$ -test** (unequal-variance), testing whether the mean final F1 differs between the two partition counts. The number of runs and the F1 variance both differ across configurations, which is why the unequal-variance form is used. This  $p$ -value determines the significance flag at  $\alpha = 0.05$ .
- **Mann–Whitney  $U$  test** (two-sided), a rank-based non-parametric check that does not assume normally distributed F1 scores. Agreement with the  $t$ -test indicates that the conclusion does not depend on the normality assumption; disagreement is a signal that more runs are needed.
- **Cohen’s  $d$  effect size**,  $d = (\bar{f}_b - \bar{f}_a)/s_p$  with  $s_p$  the pooled sample standard deviation, so that  $d > 0$  means the larger partition count has the higher mean F1. It is reported because, at small sample sizes, a non-significant  $p$ -value may reflect low statistical power rather than genuine equivalence:  $|d| \lesssim 0.2$  indicates a practically negligible difference and  $|d| \gtrsim 0.8$  a large one.

Table 17: Pairwise significance tests on the per-run final-iteration reconstruction F1 across partition counts,  $RF = 1$ . Runs per configuration: CITESEER 11, AstroPh and DBLP 4, AS-Oregon 4 at  $P = 1$  and 3 at  $P \in \{2, 4, 8\}$ , Enron 3. A negative Cohen’s  $d$  means F1 decreases as the partition count grows.

| Dataset   | Comparison | Welch<br>$t$ -test $p$ | Mann–Whitney<br>$p$ | Cohen’s $d$ | Significant<br>( $\alpha = 0.05$ ) |
|-----------|------------|------------------------|---------------------|-------------|------------------------------------|
| AS-Oregon | P=1 vs P=2 | 0.572                  | 0.629               | −0.54       | No                                 |
|           | P=1 vs P=4 | 0.095                  | 0.114               | −1.93       | No                                 |
| AS-Oregon | P=1 vs P=8 | 0.019                  | 0.057               | −4.17       | Yes                                |
|           | P=2 vs P=4 | 0.796                  | 0.700               | −0.23       | No                                 |
| AS-Oregon | P=4 vs P=8 | 0.062                  | 0.100               | −2.13       | No                                 |
|           | P=1 vs P=2 | < 0.001                | 0.029               | −9.91       | Yes                                |
| AstroPh   | P=1 vs P=4 | < 0.001                | 0.029               | −17.86      | Yes                                |

| Dataset  | Comparison    | Welch<br>$t$ -test $p$ | Mann–Whitney<br>$p$ | Cohen’s $d$ | Significant<br>( $\alpha = 0.05$ ) |
|----------|---------------|------------------------|---------------------|-------------|------------------------------------|
| AstroPh  | P=1 vs<br>P=8 | < 0.001                | 0.029               | −27.70      | Yes                                |
| AstroPh  | P=2 vs<br>P=4 | < 0.001                | 0.029               | −7.89       | Yes                                |
| AstroPh  | P=4 vs<br>P=8 | < 0.001                | 0.029               | −9.02       | Yes                                |
| CITeseer | P=1 vs<br>P=2 | 0.006                  | 0.005               | −1.42       | Yes                                |
| CITeseer | P=1 vs<br>P=4 | 0.365                  | 0.358               | +0.40       | No                                 |
| CITeseer | P=1 vs<br>P=8 | < 0.001                | < 0.001             | +4.29       | Yes                                |
| CITeseer | P=2 vs<br>P=4 | < 0.001                | < 0.001             | +3.83       | Yes                                |
| CITeseer | P=4 vs<br>P=8 | < 0.001                | < 0.001             | +5.24       | Yes                                |
| DBLP     | P=1 vs<br>P=2 | < 0.001                | 0.029               | +10.80      | Yes                                |
| DBLP     | P=1 vs<br>P=4 | < 0.001                | 0.029               | +12.66      | Yes                                |
| DBLP     | P=1 vs<br>P=8 | 0.005                  | 0.029               | −3.65       | Yes                                |
| DBLP     | P=2 vs<br>P=4 | 0.029                  | 0.057               | +2.04       | Yes                                |
| DBLP     | P=4 vs<br>P=8 | < 0.001                | 0.029               | −26.00      | Yes                                |
| Enron    | P=1 vs<br>P=2 | < 0.001                | 0.100               | +13.56      | Yes                                |
| Enron    | P=1 vs<br>P=4 | 0.040                  | 0.100               | +3.80       | Yes                                |
| Enron    | P=1 vs<br>P=8 | 0.002                  | 0.100               | +12.63      | Yes                                |
| Enron    | P=2 vs<br>P=4 | 0.323                  | 0.100               | +1.05       | No                                 |
| Enron    | P=4 vs<br>P=8 | 0.118                  | 0.200               | +1.84       | No                                 |

The tests confirm the dataset-dependent picture drawn from the mean scores. On **AstroPh**, every comparison is significant under both tests with very large negative effect sizes ( $d$  between  $-7.9$  and  $-27.7$ ): F1 degrades monotonically and unambiguously with each doubling of the partition count. On **AS-Oregon**, the degradation is milder and only the  $P = 1$  vs  $P = 8$  comparison reaches

significance under the  $t$ -test ( $p = 0.019$ ,  $d = -4.2$ ), with the Mann–Whitney test borderline ( $p = 0.057$ ) given only three runs per configuration; the direction is consistently negative but the individual per-step differences are within noise. On **CITeseer** and **Enron**, the significant comparisons are predominantly *positive*: CITeseer dips slightly from  $P = 1$  to  $P = 2$  ( $d = -1.4$ ) and then rises strongly ( $d = +5.2$  from  $P = 4$  to  $P = 8$ ), and Enron increases at every step (significant up to  $P = 4$ ,  $d = +13.6$  for  $P = 1$  vs  $P = 2$ ). **DBLP** is a mixed case rather than a clearly positive one: it rises significantly through  $P = 4$ , but then reverses sharply — the  $P = 4$  vs  $P = 8$  comparison ( $d = -26.0$ ) is the single largest-magnitude effect size in the entire table, larger than any AstroPh comparison, and  $P = 1$  vs  $P = 8$  is also a significant decrease ( $d = -3.65$ ), so DBLP ends up significantly *below* its own  $P = 1$  baseline at the largest partition count tested even though its trajectory up to  $P = 4$  is positive. AstroPh remains the only dataset whose decline is monotone across the *entire*  $P = 1$ – $P = 8$  range; DBLP’s decline is confined to its last step, but in isolation it is larger than any single AstroPh comparison.

Two caveats apply. The samples are small (as few as three runs per configuration), so the Mann–Whitney test has little power: on Enron it cannot reach  $p < 0.05$  even where the  $t$ -test and a very large  $d$  jointly indicate a clear effect. The same floor effect works the other way for AstroPh and DBLP ( $n = 4$  per group): the two-sided Mann–Whitney  $p$ -value is bottomed out at  $2/\binom{8}{4} \approx 0.029$  for any pair with no overlap, so for these two datasets every “Yes” in the Mann–Whitney column mainly confirms non-overlap rather than adding evidence beyond the  $t$ -test. In addition, the comparisons are performed without correction for multiple testing; applying a Bonferroni correction across the five comparisons per dataset, or an omnibus test (ANOVA or Kruskal–Wallis) across all partition counts at once, would be the stricter alternative. The effect sizes are reported alongside the  $p$ -values precisely so that a non-significant result is not misread as evidence of equivalence.

## Random partitioner baseline

To isolate how much of the neighbor-based partitioner’s embedding quality comes from preserving neighborhood locality, we repeated the buffered dynnode2vec experiment with a *random partitioner* that assigns each vertex to one of the  $P$  partitions uniformly at random, ignoring its neighbors entirely. All other settings (buffer size 1000,  $RF = 1$ , and the dataset-specific embedding hyperparameters listed in the partitioning hyperparameters section) are unchanged. Results for CITeseer and AstroPh, averaged over 3 iterations of the buffered stream, are reported below.

Table 18: F1 reconstruction score, edge cut, and balance for CITESEER with a random partitioner,  $RF = 1$ .

| P | F1                  | Edge cut | Balance |
|---|---------------------|----------|---------|
| 2 | $0.03\% \pm 0.02\%$ | 50.01%   | 1.02    |
| 4 | $0.03\% \pm 0.03\%$ | 70.89%   | 1.32    |
| 8 | $0.04\% \pm 0.03\%$ | 81.58%   | 1.61    |

Table 19: F1 reconstruction score, edge cut, and balance for AstroPh with a random partitioner,  $RF = 1$ .

| P | F1                  | Edge cut | Balance |
|---|---------------------|----------|---------|
| 2 | $0.04\% \pm 0.02\%$ | 49.98%   | 1.00    |
| 4 | $0.03\% \pm 0.02\%$ | 70.86%   | 1.31    |
| 8 | $0.02\% \pm 0.01\%$ | 81.43%   | 1.60    |

Compared with the neighbor-based partitioner, which reaches F1 scores of roughly 45–55% on CITESEER and 46–55% on AstroPh at the same partition counts, random assignment collapses reconstruction quality to essentially zero, with F1 below 0.05% in every configuration. This confirms that the neighbor partitioner’s F1 scores are driven by preserving neighborhood locality across partitions rather than by the embedding algorithm alone: once locality is destroyed, the per-partition embeddings become mutually incompatible and the averaged global embedding reconstructs the graph no better than chance. Edge cut is correspondingly far higher under random assignment (50–82%) than under the neighbor partitioner (23–65%), and closely tracks the value  $(P-1)/P$  expected when edge endpoints are placed independently and uniformly at random (50%, 75%, and 87.5% for  $P = 2, 4, 8$  respectively). Notably, the neighbor partitioner attains competitive embedding quality while keeping edge cut well below this random baseline, indicating that it genuinely concentrates each vertex’s neighborhood within its assigned partition.

Two aspects of this control experiment depart from the idealized theory and are worth flagging rather than glossing over. First, edge cut at  $P = 8$  (81.6% CITESEER, 81.4% AstroPh) falls noticeably short of the  $(P-1)/P = 87.5\%$  expected under fully independent uniform assignment of edge endpoints, suggesting some residual correlation between an edge’s two endpoints’ partitions even under random assignment — plausibly because a vertex’s partition, once assigned, is fixed for all its future edges rather than being re-rolled per edge. Second, balance under the random partitioner (1.32/1.61 at  $P = 4/8$  for CITESEER, 1.31/1.60 for AstroPh) is, counter-intuitively, *worse* than under the neighbor-based partitioner at the same  $P$  (e.g. 1.11/1.26 for AstroPh, in the main results table in the embedding quality, partition balance and edge cut section), even though

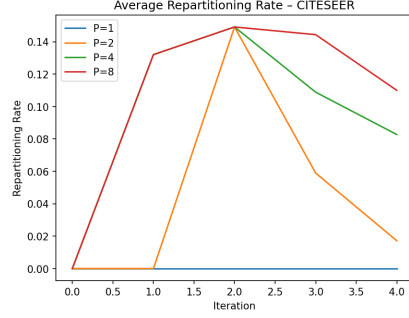
uniform-random assignment would naively be expected to balance partition sizes better than a neighbor-affinity heuristic. This suggests the random baseline is not equivalent to a single global uniform shuffle in practice — most plausibly because each vertex is assigned once, uniformly at random, at the buffer in which it first arrives, with no capacity-penalty correction, so early random imbalances across buffers can compound rather than average out. The random-partitioner numbers should therefore be read as a strong qualitative control (neighborhood locality is what drives embedding quality) rather than as a precise match to the uniform-random assignment theory.

### Effect of replication factor

Increasing the replication factor  $RF$  allows a vertex to be embedded independently in more than one partition, with the final embedding obtained by averaging the per-partition embeddings as described in the Embedding model section. This trades additional computation and storage for a more informed, less committal partition assignment. In a smaller exploratory hyperparameter sweep on CITESEER with  $P = 4$ , buffer size 1000, and capacity penalty  $\mu = 1$  (run separately from, and not directly comparable to, the main results table in the embedding quality, partition balance and edge cut section, which used a longer run), increasing the replication factor from  $RF = 1$  to  $RF = 3$  raised the F1 reconstruction score from 44.35% to 69%, indicating that hedging the partition assignment across multiple partitions can substantially mitigate the quality loss introduced by early, uncertain assignment decisions. This comes at the cost of up to  $RF \times$  the embedding computation per vertex, illustrating a quality/computation trade-off: assigning a vertex to a single partition is cheaper but riskier, while replicating it across several partitions is more expensive but more robust to a suboptimal initial assignment. Because this observation comes from a single, small-scale exploratory run, it should be read as preliminary evidence rather than a robust result; a systematic replication factor sweep across datasets and partition counts is left to future work.

### Temporal evolution

Beyond aggregate scores, it is informative to track how the repartitioning rate evolves as the event stream is ingested buffer by buffer, broken down by number



of partitions, for each of the five datasets.



Figure 8: DBLP

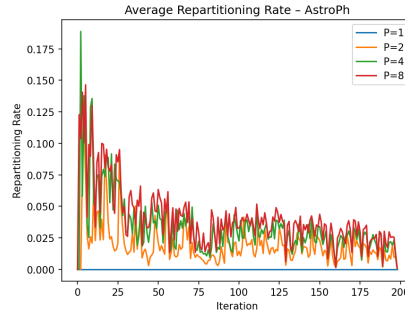


Figure 9: AstroPh

The fraction of vertices repartitioned per buffer decays rapidly after an initial warm-up phase and stays below 20% across all datasets and partition configurations. This indicates that, once the partitioner has seen enough of a vertex’s neighborhood, its assignment stabilizes and is only revised when the vertex’s local structure changes significantly, keeping the overhead of maintaining partition assignments low throughout the stream.

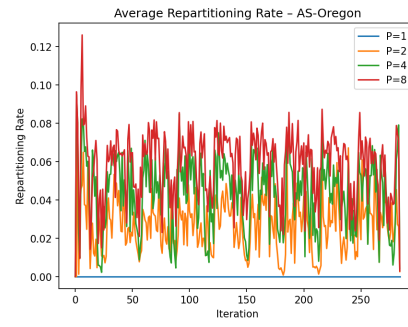


Figure 10: AS-Oregon

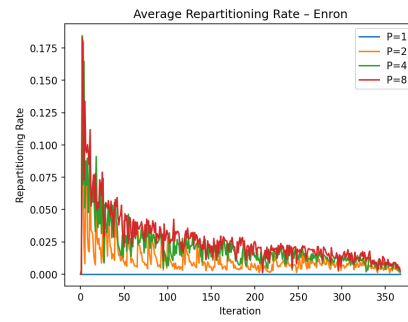


Figure 11: Enron

## Conclusion

In this paper, we studied the trade-off between early and informed partition assignment in distributed dynamic graph embedding, in a setting where a distributed dynnode2vec-style embedding model processes graph events in buffered batches. We described a neighbor-based partitioning strategy with a tunable replication factor and capacity penalty, and used it to empirically characterize how partition count and replication factor affect embedding quality, partition balance, edge cut, and repartitioning stability across five real-world dynamic graph datasets.

The results show that the impact of partitioning on embedding quality is dataset-dependent, though not a clean three-vs-two split. AstroPh degrades monotonically and significantly with every doubling of  $P$  (Cohen’s  $d$  from  $-7.9$  to  $-27.7$ ), and AS-Oregon also trends downward, though only its  $P = 1$  vs  $P = 8$  gap is statistically distinguishable from run-to-run noise. CITESEER and Enron are stable or improve significantly across most of the partition range. DBLP is the intermediate case: it improves significantly up to  $P = 4$  but then reverses sharply, ending significantly below its own  $P = 1$  baseline at  $P = 8$  with the single largest-magnitude effect size measured anywhere in this study ( $d = -26.0$  for  $P = 4$  vs  $P = 8$ ) — partitioning does not help DBLP uniformly, only up to a point. Pairwise Welch  $t$ -tests and Mann–Whitney  $U$  tests on the per-run final F1 scores, together with Cohen’s  $d$  effect sizes, back this characterization, with the caveat that the very small within-configuration standard deviations observed (e.g. AstroPh  $\pm 0.11\%$  at  $P = 1$ ) mean that even modest percentage-point differences register as statistically significant with large effect sizes; the percentage-point magnitude, not the  $p$ -value or  $d$  alone, is what should carry the practical interpretation. Of the dataset properties measured (density, clustering coefficient, modularity), graph density correlates most closely with this split, though it does not fully explain it, since no single threshold on density, clustering coefficient, or modularity cleanly separates the two groups; identifying the underlying driver is left to future work. On most datasets, the partitioner keeps balance close to the ideal value, though CITESEER shows a more pronounced imbalance at higher partition counts; across all datasets and partition counts it keeps the repartitioning rate low and decaying after an initial warm-up period, indicating that a simple neighbor-based, buffer-driven partitioning strategy is a practical and largely, though not universally, stable choice for online graph partitioning. Partitioning time itself is consistently smaller than embedding time — directly measured at roughly 20–30% of embedding time at the buffer size used throughout this study, and an even smaller share at lower buffer sizes — confirming that the dominant systems bottleneck in this setting is the embedding computation on the largest or most active partition rather than the assignment decision itself.

The replication factor experiment provides preliminary evidence for the trade-off that motivates this paper: in a single exploratory configuration, hedging a vertex’s partition assignment across multiple partitions and averaging the resulting

embeddings substantially mitigated the quality loss caused by early, uncertain assignment decisions, at the cost of additional computation proportional to the replication factor. This suggests that systems which must assign vertices to partitions immediately, with little information about their eventual neighborhood, benefit from replication as a way to hedge against a suboptimal initial assignment, while systems that can tolerate additional latency may instead delay assignment until more information is available, avoiding the replication overhead altogether. Together, these findings indicate that neither immediate nor delayed assignment is uniformly preferable; the right choice depends on the graph’s community structure and on which resource, latency or computation, is more constrained in a given deployment.

As future work, a more direct comparison between delayed and immediate assignment strategies is warranted, explicitly measuring assignment latency alongside embedding quality, rather than using the replication factor solely as a proxy for the trade-off. Extending the replication factor study beyond a single dataset and partition count would also strengthen the empirical evidence presented here. Finally, evaluating the resulting embeddings on downstream tasks such as link prediction and node classification, in addition to the reconstruction F1 score used in this paper, would help establish whether the observed trends generalize beyond graph reconstruction.

## References

- Adamic, Lada A, and Natalie Glance. 2005. “The Political Blogosphere and the 2004 US Election: Divided They Blog.” *Proceedings of the 3rd International Workshop on Link Discovery*, 36–43.
- Albert, Réka, and Albert-László Barabási. 2002. “Statistical Mechanics of Complex Networks.” *Reviews of Modern Physics* 74 (1): 47.
- Backstrom, Lars, Dan Huttenlocher, Jon Kleinberg, and Xiangyang Lan. 2006. “Group Formation in Large Social Networks: Membership, Growth, and Evolution.” *Proceedings of the 12th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 44–54.
- Barabási, Albert-László, and Réka Albert. 1999. “Emergence of Scaling in Random Networks.” *Science* 286 (5439): 509–12.
- Benlic, Una, and Jin-Kao Hao. 2010. “An Effective Multilevel Memetic Algorithm for Balanced Graph Partitioning.” *2010 22nd IEEE International Conference on Tools with Artificial Intelligence* 1: 121–28.
- Çatalyürek, Ümit, Karen Devine, Marcelo Faraj, et al. 2023. “More Recent Advances in (Hyper) Graph Partitioning.” *ACM Computing Surveys* 55 (12):

1–38.

Church, Kenneth Ward. 2017. “Word2Vec.” *Natural Language Engineering* 23 (1): 155–62.

Fang, Peng, Arijit Khan, Siqiang Luo, et al. 2023. “Distributed Graph Embedding with Information-Oriented Random Walks.” *arXiv Preprint arXiv:2303.15702*.

Grover, Aditya, and Jure Leskovec. 2016. “Node2vec: Scalable Feature Learning for Networks.” *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 855–64.

Ji, Genlin, and Xiaohan Ling. 2007. “Ensemble Learning Based Distributed Clustering.” *Emerging Technologies in Knowledge Discovery and Data Mining: PAKDD 2007 International Workshops Nanjing, China, May 22–25, 2007 Revised Selected Papers 11*, 312–21.

Krebs, Valdis E. 2002. “Mapping Networks of Terrorist Cells.” *Connections* 24 (3): 43–52.

Leskovec, Jure, Daniel Huttenlocher, and Jon Kleinberg. 2010a. “Predicting Positive and Negative Links in Online Social Networks.” *Proceedings of the 19th International Conference on World Wide Web*, 641–50.

Leskovec, Jure, Daniel Huttenlocher, and Jon Kleinberg. 2010b. “Signed Networks in Social Media.” *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 1361–70.

Leskovec, Jure, Kevin J Lang, Anirban Dasgupta, and Michael W Mahoney. 2009. “Community Structure in Large Networks: Natural Cluster Sizes and the Absence of Large Well-Defined Clusters.” *Internet Mathematics* 6 (1): 29–123.

Li, Mu, Li Zhou, Zichao Yang, et al. 2013. “Parameter Server for Distributed Machine Learning.” *Big Learning NIPS Workshop* 6.

Lombardo, Gianfranco, and Agostino Poggi. 2019. “A Scalable and Distributed Actor-Based Version of the Node2Vec Algorithm.” *Proceedings of the 20th Workshop From Objects to Agents* (Parma, Italy), 134–41.

Mahdavi, Sedigheh, Shima Khoshraftar, and Aijun An. 2018. “Dynnode2vec: Scalable Dynamic Network Embedding.” *2018 IEEE International Conference on Big Data (Big Data)*, 3762–65.

Romero Ruiz, Emmanuel, and Carlos Segura. 2018. “Memetic Algorithm with

- Hungarian Matching Based Crossover and Diversity Preservation.” *Computación y Sistemas* 22 (2): 347–61.
- Rozemberczki, Benedek, and Rik Sarkar. 2021. *Twitch Gamers: A Dataset for Evaluating Proximity Preserving and Structural Role-Based Node Embeddings*.
- Sanders, Peter, and Christian Schulz. 2011. “Engineering Multilevel Graph Partitioning Algorithms.” *European Symposium on Algorithms*, 469–80.
- Sanders, Peter, and Christian Schulz. 2012. “Distributed Evolutionary Graph Partitioning.” *2012 Proceedings of the Fourteenth Workshop on Algorithm Engineering and Experiments (ALENEX)*, 16–29.
- Savić, Miloš, Mirjana Ivanović, and Bojana Dimić Surla. 2017. “Analysis of Intra-Institutional Research Collaboration: A Case of a Serbian Faculty of Sciences.” *Scientometrics* 110: 195–216.
- Stanton, Isabelle, and Gabriel Kliot. 2012. “Streaming Graph Partitioning for Large Distributed Graphs.” *Proceedings of the 18th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 1222–30.
- Tsourakakis, Charalampos, Christos Gkantsidis, Bozidar Radunovic, and Milan Vojnovic. 2014. “Fennel: Streaming Graph Partitioning for Massive Scale Graphs.” *Proceedings of the 7th ACM International Conference on Web Search and Data Mining*, 333–42.
- Ugander, Johan, and Lars Backstrom. 2013. “Balanced Label Propagation for Partitioning Massive Graphs.” *Proceedings of the Sixth ACM International Conference on Web Search and Data Mining*, 507–16.
- Van Der Hofstad, Remco. 2024. *Random Graphs and Complex Networks*. Vol. 54. Cambridge university press.
- Watts, Duncan J, and Steven H Strogatz. 1998. “Collective Dynamics of ‘Small-World’ networks.” *Nature* 393 (6684): 440–42.
- Yip, Hong Yung, Chidaksh Ravuru, Neelabha Banerjee, et al. 2023. *RESTORE: Graph Embedding Assessment Through Reconstruction*. <https://arxiv.org/abs/2308.14659>.
- Zachary, Wayne W. 1977. “An Information Flow Model for Conflict and Fission in Small Groups.” *Journal of Anthropological Research* 33 (4): 452–73.